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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Third Semester B.E. (E&C) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21EC303/17EC406(20EC302) – DIGITAL SYSTEM DESIGN WITH VERILOG (DIGITAL SYSTEMS DESIGN)

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1. a)	Design a combinational logic truth table such that an output (Y) is high when a majority of four inputs (I_3, I_2, I_1, I_0) are true. Also obtain the simplified Boolean expression for the output and draw the logic diagram.	8	L*3	1	1
b)	Find the prime implicants, essential prime implicants and minimal product for the following expression. $T = F(w, x, y, z) = \pi M(1, 3, 8, 10, 12, 13, 14, 15)$	6	L3	1	1
c)	Explain the structure of Verilog module with the help of a Verilog code for a Full Adder.	6	L2	2	1
2. a)	Design a Full Subtractor with the help of its truth table and draw the logic diagram. Also write a Verilog code for a Full Subtractor using dataflow modeling.	8	L2	2	1
b)	Convert the following expressions into their requested canonical forms: i) $T = f(x, y, z) = xy + x\bar{z} + y\bar{x} + yz$ (SOP) ii) $P = f(p, q, r, s) = (\bar{p} + q + s)(r + \bar{s})$ (POS)	6	L3	1	1
c)	Find the prime implicants, essential prime implicants and minimal sum for the following expression. $F(a, b, c, d) = \pi M(0, 1, 4, 5, 8, 9, 11)$	6	L3	1	1
3. a)	Explain the principle of Carry Look-ahead Adder. Derive the expressions for the carry outputs of a 4-stage Carry Look-Ahead Adder and draw the block diagram.	8	L2	2	1
b)	Determine the minimal sum for the following Boolean function. $F(a, b, c, d) = \sum m(2, 3, 4, 5, 13, 15) + dc(8, 9, 10, 11)$	6	L3	1	1
c)	Write a Verilog behavioral code for a 4-bit parallel Adder/Subtractor by taking a control signal 'Mode'. If 'Mode'=1 it performs addition operation otherwise subtraction.	6	L2	2	1
Unit – II					
4. a)	Explain the operation of Master-slave JK flip flop using gated SR latches with the help of logic diagram, function table and its timing diagram.	8	L2	4	1
b)	Implement the following Boolean functions using 3:8 decoder with active high outputs minimizing the number of inputs to the gates $F1(a, b, c) = \sum m(2, 3, 4, 5, 6)$ $F2(a, b, c) = \pi M(2, 3, 7)$	6	L3	3	1
c)	Write a Verilog code for an 8:1 Multiplexer using 'case' statement in behavioral modeling. Also write its truth table.	6	L2	3	1
5. a)	Explain the 0's catching and 1's catching problem in Master slave JK flip flop.	8	L2	4	1

P.T.O.

- b) Design a 1-bit Full Adder using 4x1 Multiplexers and external gates if required, using suitable procedure.
- c) Write a Verilog code for a T flip flop in behavioral modeling with reset signal. Also obtain its characteristic equation.
6. a) Explain the working of a positive edge triggered D flip flop with the help of logic circuit and truth table.
- b) Explain the operation of a gated SR latch constructed using NAND gates only with the help of logic diagram, symbol and function table.
- c) With the help of truth table write a Verilog behavioral code for a 4:2 Encoder using if-else statements.

6	L3	3	1
6	L2	4	1
8	L2	4	1
6	L2	4	1
6	L2	3	1

Unit – III

7. a) Construct the excitation table, transition table, state table and state diagram for the synchronous sequential circuit shown in Fig. 7(a).

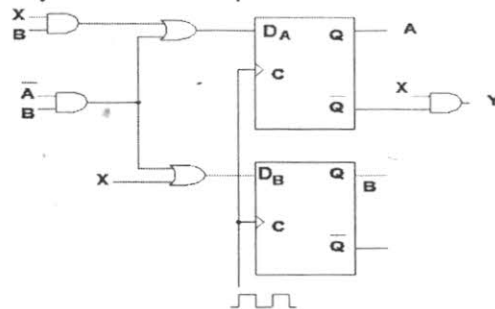


Figure. 7(a)

10	L3	5	1
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- b) Design a Universal shift register incorporating four Multiplexers and four positive edge triggered D flip flops, having the behavior specified in the table given below.

Select Lines a1 a0	Register operations
0 0	Parallel Load
0 1	Hold
1 0	Complement contents
1 1	Circular shift right

5	L3	5	1
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- c) Write a Verilog code for a 5-bit synchronous updown counter with active high reset and control signal 'up_down' in behavioral modeling. If 'up_down' =1, upcounting is performed else down counting.
8. a) Design a synchronous counter to count the sequence 0->4->1->2->6->0 using positive edge triggered D flip flops.
- b) Design a 4-bit binary ripple up counter using negative edge triggered T-flip flop. Also write the counting sequence.
- c) The sequential circuit depicted by the state diagram in Figure 8. (c) is a two-bit binary counter controlled by input 'x_in'. The output 'y_out' becomes 'high' when the count reaches binary 11. Write a Verilog code for the sequential circuit in behavioral modeling.

5	L2	5	1
10	L3	5	1
5	L3	5	1

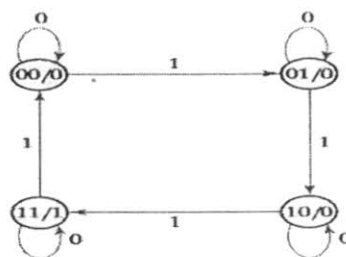


Figure. 8(c)

5	L2	5	1
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